

UEXA: Self-Acting Against Extreme Climate Risks

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UEXA is a \$4M, three-year research program to prevent climate-driven disaster deaths through two integrated pillars: PREDICT and RESPOND. PREDICT is a “data-assimilative” “uncertainty-aware” “cross-lifeline” digital twin that anticipates which water, power, road, communication, and medical lifelines will fail before and during the disaster. RESPOND solves an open operations-research problem: how to allocate autonomous sensing-and-response under simultaneous distributional shift and observability collapse, so that high-risk unfound populations can be located within a 3-hour decision-to-detection window, not a 72-hour relief horizon.

What am I trying to do?

Climate disasters increasingly kill not only through their direct physical hazard, but through cascading failures of the lifelines people depend on to survive. Floodwater, landslides, and storms are the visible trigger; the deeper risk emerges when critical infrastructure (cf. water, power, roads, communications, and medical systems) fail together, delay people’s evacuation, interrupt care, isolate communities, and prevent responders from seeing where help is still needed.

The 2019 East Japan Typhoon (Hagibis) exposed this gap starkly. 65% of the 84 deaths were people aged 65 or older, and 79% were older adults among those who died at home. These figures point not only to the physical hazard of flooding, but also to persistent barriers to timely and supported evacuation for older people and others requiring assistance. In the same event, Soma Central Hospital in Fukushima prefecture nearly lost its ability to continue hemodialysis after the city’s water supply was destroyed; treatment continued only because the Self Defence Force managed to deliver water by truck in time. The system held by improvisation, not by design, and not every disaster will be that lucky.

So far, these cascading loops of infrastructure failure have not been treated as a computable, continuously updated operational problem. Existing disaster systems often model the hazard, the exposed population, or individual infrastructure assets, but they rarely model how failure in one lifeline propagates through others and turns into human mortality risk, both in the planning phase as well as in real time during the disaster. Prediction and response are also built by separate communities, on fragmented abstractions that never fully feed each other. This limitation is also amplified in the Global South, where the data infrastructure for high-quality digital twins is sparse or absent.

The objective of this program UEXA is therefore twofold:

1. **PREDICT:** Build a data-assimilative, uncertainty-aware digital twin that, hours to days before a climate disaster strikes, identifies which critical lifelines are likely to fail, how those failures may cascade across infrastructure systems, and how human mobility dynamics will determine which populations become isolated, unable to evacuate, or difficult to rescue.
2. **RESPOND:** Solve the operations-research problem of how to allocate a heterogeneous autonomous sensing-and-response fleet under non-stationary disaster conditions and degrading observability, so that the system can continuously update the digital twin, close information gaps, and locate high-risk unfound populations within a 3-hour decision-to-detection window.

Together, the two pillars target a compression of the disaster see-decide-act loop **from a conventional 72-hour relief horizon to a 3-hour decision-to-detection target**, keeping the landslide-buried cohort within the 1-6 hour survival window and the flood-stranded or community-isolated cohorts within the 24-72 hour survival window.

How is it done today?

Current approaches to climate-disaster response remain fragmented across two dimensions that this UEXA program addresses, predictive cross-domain lifeline modeling, and real-time decision-making under degrading observability.

On the prediction side, lab-scale work has begun to converge toward urban foundation models with mobility, semantic context, and intervention simulation. Nation-wide 2D/3D building auto-update from satellite imagery is emerging as the geospatial substrate. Single-domain systems do work, but no integrated system jointly predicts the failure of water, power, road, and medical lifelines under a unified climate-scenario model, and so none can pre-position rescue against a joint prediction.

On the response side, individual instruments are no longer the bottleneck. Multimodal drones, satellite tasking, mobile sensors, and crowd-sourced channels are technologically mature. What is missing is the orchestration layer that allocates this heterogeneous fleet against a coherent belief state under hard real-world constraints. Today these decisions rely on static priors and scheduling heuristics that never close the loop with an evolving world.

Under disaster conditions the system enters distributional shift (mobility, infrastructure, prior models all break) and observability collapse (sensors degrade, comms drop) simultaneously. No current framework jointly addresses predictive cross-domain lifeline modeling, data scarcity, dynamic disruption, real-time human mobility, and optimal autonomous-sensing allocation. UEXA proposes to close all of these together.

What is new in UEXA's approach?

UEXA organizes around two pillars, PREDICT and RESPOND, which are distinct in function but tightly integrated in operation. The program is intentionally asymmetric; Pillar 1 is critical enabling work with bounded research extensions, while Pillar 2 is the major ARPA-hard scientific challenge.

Pillar 1 PREDICT builds the cross-lifeline digital twin: the continuously updated foundation that anticipates how infrastructure failures, human mobility dynamics, and population vulnerability interact before and during climate disasters. This twin has standalone value for preparedness, evacuation planning, and infrastructure-risk management. It also provides the operational state, uncertainty surface, and failure-cascade priors that Pillar 2 requires.

Pillar 2 RESPOND then addresses the major ARPA-hard scientific problem of the program: how to allocate autonomous sensing and response under simultaneous distributional shift and observability collapse. It is structured as the UEXA Challenge, a benchmarked competition to discover and down-select solution families, not only as an internal workstream. In other words, UEXA does not treat prediction and response as separate sequential tasks, but it builds a closed loop in which prediction guides sensing and response, while sensing and response continuously update the twin as the disaster evolves.

Pillar 1 - PREDICT: A Data-Assimilative, Uncertainty-Aware Digital Twin

Pillar 1 constructs a continuously updated digital twin of a region's critical infrastructure, including water, power, roads, medical services, and communications, and uses it to predict which lifelines will fail under a given climate scenario. The twin models lifelines and their couplings as one interacting system, so that a failure cascade is predicted as a single operational phenomenon, not as a set of disconnected events. (e.g., a power outage may disable hospital dialysis; road disruption may prevent water delivery; communication failure may delay evacuation support for older people or medically dependent populations) The core object of prediction is therefore not infrastructure failure alone, but the transformation from lifeline failure into human mortality risk.

The substrate combines auto-updated 2D/3D building geometry from satellite imagery, a mobility-aware human-centric population state, multimodal data assimilation fusing structured signals (remote sensing and infrastructure sensors) with unstructured signals (social media, ground imagery, responder reports), and a spatially resolved uncertainty surface that Pillar 2 consumes. This allows UEXA to estimate not only where the hazard is, but which lifelines are degrading, which populations are becoming isolated or unable to evacuate, and where uncertainty is highest.

Three extensions are required for cross-lifeline disaster operation: cross-domain coupling modeling to capture failure propagation between lifelines; cross-city transferability under data sparsity and uncertainty, especially for Global South deployment where ground-truth records are limited; and intervention-aware tokenization so evacuations, infrastructure failures, temporary access routes, ad-hoc shelters, and emergency service disruptions can be injected as first-class context. These are real research challenges, but they are bounded extensions of mature research lines. The contribution of Pillar 1 is to unify them into an operational-scale, uncertainty-aware digital twin that can serve as the substrate for real-time sensing and response.

Pillar 2 - RESPOND: the UEXA Challenge

Given the continuously updated cross-lifeline twin developed in Pillar 1, Pillar 2 runs in real time during the disaster, when prediction is incomplete or the world has evolved beyond the model. Its task is to allocate a heterogeneous autonomous fleet against the remaining unfound population, under conditions where the world distribution and the observation channel are both non-stationary. This is the open scientific problem the program is built around. Formally:

$$\max_{A \in \Pi} E [I (X^* ; Y | A, \omega)]$$

X^* is the unfound human-exposure state; Y is observations from the deployed fleet; A is the sensing allocation policy (where, when, what platform, what payload) drawn from a feasible set Π encoding real-world constraints; ω is the non-stationary world state. The objective: maximize expected information about the unfound.

The problem is hard because the world distribution $p(\omega)$ shifts (mobility deviates, infrastructure fails, priors become unreliable) while simultaneously the observation channel $p(Y | A, \omega)$ fades (sensors damaged, comms degrading, allocation feasibility changing by the minute). Existing literature treats these regimes separately; UEXA's bet is they must be solved together.

Two developments make UEXA's bet plausible “now”;

1. recent AI capability (cf. multimodal foundation models, high-resolution physical-state surrogates, and generative data assimilation) enables updating high-dimensional disaster, infrastructure, and human-exposure states, even from sparse, noisy, and heterogeneous observations, which directly addresses the observation-collapse half of the problem.
2. The program's mathematical core is a control-theoretic reframing of the observation channel itself. While standard approaches treat degraded observations passively, in UEXA, the observation collapse itself becomes part of the state to be estimated and controlled; where is observation faded, why, and by how much? Once those become latent variables, the allocation policy A no longer just chooses where to look; it chooses where to restore observability, bypassing or complementing the degraded channels. That reframing is the program's distinctive theoretical contribution, and what makes the joint shift $\times$ collapse problem newly tractable.

In summary, Pillar 1 reduces the size and uncertainty of the rescue problem before impact, while Pillar 2 collapses the remaining uncertainty during response. Integrated through a shared uncertainty representation and decision basis, they fit the see-decide-act loop inside the 3-hour target across major high-risk cohorts.

Program Technical Gates

The central technical objective of UEXA is to compress the disaster see-decide-act loop from the current 72-hour relief horizon to a 3-hour decision-to-detection target. Achieving this requires two linked breakthroughs. Before disaster impacts, Pillar 1 must reduce the number and spatial uncertainty of people who will require rescue by predicting cascading lifeline failure, mobility constraints, and population isolation. During the disaster, Pillar 2 must reduce the remaining uncertainty fast enough to locate and prioritize the unfound population within hours.

The two pillars use different execution models. Pillar 1 will be developed by a dedicated core team responsible for the cross-lifeline twin, data assimilation, mobility-aware population state, and spatial uncertainty representation. Pillar 2 will be run as the UEXA Challenge: a competition-based research mechanism to discover, compare, and down-select algorithms.

By the end of Year 1, the program should establish a reproducible benchmark environment. Pillar 1 should produce an event-scale cross-lifeline twin for one historical disaster and demonstrate that it outperforms single-domain baselines in predicting lifeline failure and exposed population states. Pillar 2 should produce a peer-reviewable UEXA Challenge specification, including the mathematical formulation, operational constraints, baseline allocation policies, and scoring rules.

By the end of Year 2, the program should demonstrate bounded technical performance in simulation. Pillar 1 should predict coupled failure cascades across multiple lifelines under non-stationary disaster conditions. Pillar 2 should produce candidate algorithms that outperform baseline search-and-allocation policies under observation-channel degradation. Target metrics should include:

Pillar 1 = reducing the high-risk search area by at least 70% before response begins; achieving calibrated cross-lifeline failure prediction under synthetic compound disasters

Pillar 2 = improving time-to-detection by at least 3x against static or greedy allocation baselines; and maintaining performance under partial sensor, communication, and mobility loss.

By the end of Year 3, the program should demonstrate an integrated closed-loop technical capability in counterfactual replay. The Pillar 1 twin should provide the evolving belief state, uncertainty surface, and failure-cascade priors, and Pillar 2 should allocate sensing and response assets; and each new observation should re-enter the twin and change the next decision cycle. The technical target is to show, in at least two replayed climate disasters, that the integrated system can reduce the expected time to locate high-risk unfound populations from a 72-hour relief horizon to a 3-hour decision-to-detection window under defined disaster scenarios.

The architecture will be stress-tested on selected Japanese climate-driven disasters, like below for examples, each chosen to expose a different technical failure mode:

- 1. The 2015 Kanto-Tohoku flood** (Kinugawa breach, 4,000+ stranded): predict the unforeseen levee failure; locate the unfound stranded population in hours, not days.
- 2. The 2018 West Japan heavy rain** (Mabi hospital flooded, 150 patients with treatment interrupted): predict the medical lifeline collapse; route rescue ahead of human dispatch.
- 3. The 2019 East Japan Typhoon (Hagibis)** (¥1.8T damage, 135 levees breached, 22K homes without power and 133K without water for over two weeks): predict joint multi-lifeline failure across Kanto; operate across multiple disaster modes at once.

What's the impact if successful?

UEXA shifts disaster response from delayed assessment to a continuously running prediction-to-decision system. The pre-disaster twin enables cross-domain infrastructure hardening and vulnerable-population pre-evacuation that no current Japanese system performs at scale; the real-time OR layer enables autonomous fleet allocation that no current civilian system performs at all.

Compressing the response loop from 72h to 3h materially increases the share of high-risk cohorts reached within survival-relevant windows, rather than implying that every person can always be saved. The same compression generates substantial second-order economic returns: fewer SDF and helicopter hours per rescue, shorter evacuation periods, less time during which hospitals operate at degraded capacity, less time during which businesses and supply chains run in disruption mode. The difference is between climate disaster as a recurring, compounding drag on national productivity and a more contained, measurable risk.

The two-pillar architecture is portable beyond climate disasters. The cross-lifeline predictive twin extends to non-climate infrastructure crises such as large-scale blackouts, pandemic response, conflict-zone humanitarian operations. The UEXA Challenge formulation extends to any domain orchestrating mature autonomous tools against a non-stationary world.

At scale, UEXA is deliberately built to operate where conventional data infrastructure is limited, making it adaptable to Global South settings where most climate disasters strike and where resource constraints on rescue assets are tightest. The marginal cost of extending UEXA to a new Global South country should be far below the cost of building conventional response infrastructure from scratch.

Budget

The program is scoped at \$4.0M over three years. The structure reflects the two-pillar division and reserves significant integration and validation funding.

By pillar

- **Pillar 1 PREDICT: Cross-lifeline twin substrate (\$1.45M).** Core team for the twin, mobility-aware exposure state, failure cascades, uncertainty surfaces, and data-sparse transfer. Target: 70%+ pre-impact search-space reduction.
- **Pillar 2 RESPOND: UEXA Challenge (\$1.55M).** Competition mechanism for benchmarks, synthetic disaster generator, baseline policies, solver awards, external evaluation, and integration grants. Target: 3x+ faster time-to-detection under shift and observability collapse.
- **Closed-loop integration and stress testing (\$0.70M).** Connect Pillar 1 states to Pillar 2 allocation policies; replay disasters; test sensor, communication, and mobility loss.
- **Program architecture and evaluation (\$0.30M).** Technical program management, partner engagement, benchmark governance, external review, documentation, and dissemination.

By year

Year	Investment	Focus
Year 1: Make computable	\$1.1M	baseline twin, replay environment, UEXA Challenge specification, scoring rules
Year 2: Produce breakthroughs	\$1.6M	coupled cascade prediction, first challenge round
Year 3: Close the loop	\$1.3M	integrate winning algorithms, run closed-loop replay, stress-test the 72-hour to 3-hour path
Total	\$4.0M	